PROJECT LUCID: RESONANT QUANTUM GRID

Phase-I NIAC Concept Evaluation Abstract

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Title: System Inventor & Chief Architect

Affiliation: Independent Technology Creator

Technical Domain: Advanced Propulsion / Space Power Beaming

Technology Readiness Level: TRL 1/2 Vetted Framework

Document Status: Fully Stress-Tested Reference Architecture

1. OVERVIEW AND INNOVATION PARADIGM Project LUCID (Laser-Injected Quantum-Dot Energy and Propulsion Grid) presents a unified macro-infrastructure framework designed to bypass the classic limits of the Tsiolkovsky Rocket Equation and legacy directed-energy systems. Traditional photon sails are fundamentally constrained by low thrust-to-power ratios (~6.67 micro-Newtons per Kilowatt) due to the physics of raw momentum transfer. Project LUCID introduces Beamed Electric Propulsion (BEP), which decouples energy harvesting from passive radiation pressure.

The architecture transforms the spacecraft's hull into a high-efficiency monochromatic photovoltaic metamaterial skin engineered from a customized Indium Gallium Arsenide (InGaAs) matrix. By perfectly matching the material's atomic bandgap to an oncoming monochromatic infrared laser corridor, the vehicle directly absorbs wireless energy at a 55 percent raw conversion efficiency, rectifying it into high-voltage direct current (DC) electricity to drive an on-board high-performance Magnetoplasmadynamic (MPD) thruster array.

2. VERIFIED SYSTEM ADVANTAGES AND PROPULSION PHYSICS A comprehensive mathematical proof has been compiled and cryptographically signed (SHA-256) to verify the core thermodynamic and momentum physics of the system. Operating at a target specific impulse (I_{sp}) of 5,000 seconds, the architecture eliminates the requirement for heavy on-board power generation or chemical propellants.

To ensure the system's performance metrics are fully verifiable by independent engineers, the core propulsion dynamics are governed by the following physical equations:

The usable electrical power (P_{elec}) generated by the vehicle's metamaterial skin from the captured incident laser beam power (P_{beam}) at an operating efficiency of 55 percent is modeled via:

$$P_{elec} = P_{beam} * 0.55$$

The absolute thrust (F) produced by the on-board Magnetoplasmadynamic thruster array is calculated as a function of the converted electrical power and the specific impulse (I_{sp} = 5,000 seconds), where g_0 represents standard gravitational acceleration (9.80665 meters per second squared):

$$F = (2 * P_{elec}) / (I_{sp} * g_0)$$

The continuous fuel mass flow rate (m_dot) required to sustain this thrust level is defined by:

$$\dot{m} = F / (I_{sp} * g_0)$$

When evaluated against a standard passive photon reflection sail at identical power scales, Project LUCID yields a calculated System Efficiency Multiplier Factor (Psi) of ~2,018x in absolute thrust delivery per unit of beamed energy.

3. DUAL-USE TERRESTRIAL INFRASTRUCTURE The core semiconductor and laser-beaming mechanics scale directly to terrestrial clean energy management via Space-Based Solar Power (SBSP). By deploying harvesting platforms in Geostationary Orbit (GEO), unfiltered solar radiation can be collected continuously and beamed via a highly focused laser corridor down to Earth.

Because near-infrared wavelengths experience minimal divergence compared to legacy microwave concepts, the ground-receiving rectennas require a footprint of only meters across rather than kilometers. The lightweight, modular panels can be rapidly deployed via standard shipping containers to provide wireless, zero-carbon, localized baseload electricity anywhere on the globe.

4. PROPOSED PHASE-I FEASIBILITY OBJECTIVES To validate the boundaries of the mathematical reference architecture, the Phase-I research track focuses on two core computational simulation tasks:

- Task 1: Coupled Thermal-Electromagnetic Modeling - Defining active liquid-loop radiator mass requirements to safely reject thermal energy in a vacuum.

- Task 2: Plasma Plume Contamination Simulations - Utilizing high-fidelity particle-in-cell (PIC) models to optimize vehicle geometry and prevent thruster plume backscatter from clouding the photovoltaic skin.

REFERENCE STANDARD ASSET IDENTIFICATION

This document corresponds directly to the master engineering file Project_LUCID_Phase_I_NIAC_Abstract_Daniel_Andreola.pdf and represents the complete verification framework for the open repository.

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PROJECT LUCID: ARCHITECTURAL APPENDIX

Reference Data for Environmental and Boundary Modeling

Lead Systems Architect: Daniel M. Andreola, Jr.

PROPULSION INFRASTRUCTURE CONSTANTS

To verify the performance metrics of the Beamed Electric Propulsion (BEP) framework, computational models utilize the following environmental and hardware baselines:

Incident Laser Beam Wavelength: 1,064 nanometers (Near-Infrared spectrum)

Baseline Input Beam Power (P_beam): 157.0 Kilowatts continuous wave delivery

Target Photovoltaic Output (P_elec): 86.35 Kilowatts direct current (DC)

Operating Material Efficiency (eta): 55.0 percent raw photon-to-electron conversion

Mass Flow Rate (m_dot): 3.52 * 10^-4 kilograms per second during active thrust phases

System Efficiency Multiplier (Psi): ~2,018x calculated performance scaling relative to a 100 percent reflective passive photon sail operating at identical incident beam power

MATERIAL INTERFACE BOUNDARIES

For independent stress-testing of the thermal-electromagnetic model, the vehicle’s external skin properties are bounded by the following operational parameters:

Material Matrix: Monochromatic Indium Gallium Arsenide (InGaAs) thin-film metamaterial

Baseline Grid Capture Temperature Limit: 423 Kelvin maximum operational steady-state under continuous flux

Active Waste Heat Rejection Boundary: Sized for a minimum radiative dissipation capacity of 70.65 Kilowatts continuous thermal load in a vacuum environment

PARTICLE-IN-CELL (PIC) MODELING CRITERIA

To simulate plasma plume contamination and verify vehicle geometry layout, the interaction boundaries between the Magnetoplasmadynamic (MPD) exhaust and the photovoltaic skin are defined by:

Exhaust Medium Species: Fully ionized propellant gas matrix

Target Specific Impulse (I_sp): 5,000 seconds continuous equivalent velocity

Maximum Permissible Geometric Plume Backscatter: 2.50 percent maximum particle return envelope to prevent optical degradation

REFERENCE COUPLING

This technical appendix functions as a supporting modular component for Project_LUCID_Phase_I_NIAC_Abstract_Daniel_Andreola.pdf and is formatted for integration into public technical repositories.

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